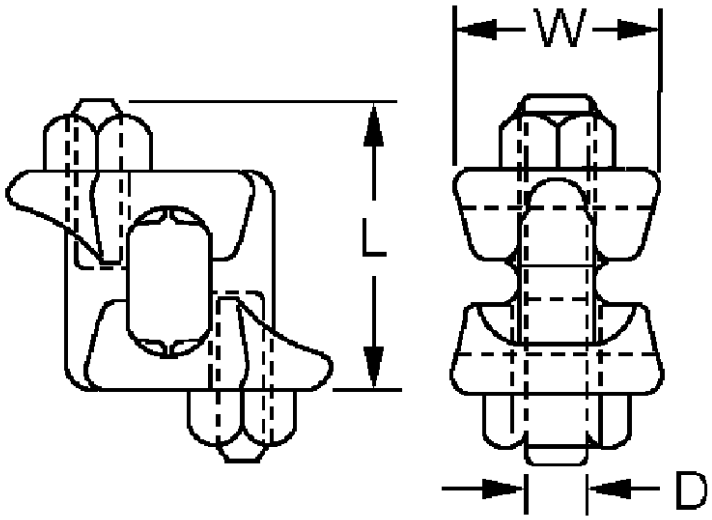
INCH-POUND
MS51868D
16 April 2012
SUPERSEDING
MS51868C
9 March 2011

DETAIL SPECIFICATION SHEET

CLAMP, WIRE ROPE, DOUBLE SADDLED, DOUBLE-GRIP, STEEL, TYPE III

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and FF-C-450.



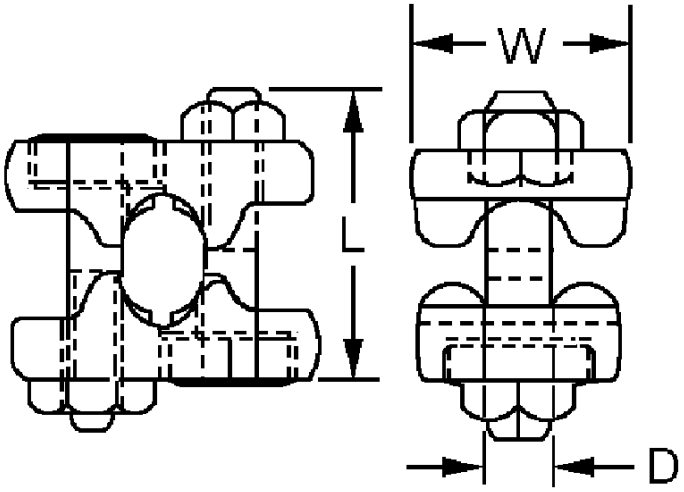
Type III, Class 1

FIGURE 1. Clamp, double grip, double saddled wire rope – type III, class 1.

AMSC N/A

FSC 4030

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Type III, Class 2

FIGURE 2. Clamp, double grip, double saddled wire rope – type III, class 2.

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TABLE I. Clamp dimensions and number of clamps used. 1/ 2/ 3/

Dash number						Clamp size	D	L
Hot dip galvanized		Electrodeposited		Mechanically deposited				
Class 1	Class 2	Class 1	Class 2	Class 1	Class 2		Thread minimum	Length minimum
1	51	101	151	201	251	3/16 (.188)	10-24	1.281
2	52	102	152	202	252	1/4 (.250)	1/4-20	1.281
3	53	103	153	203	253	5/16 (.3125)	¼-20	1.438
4	54	104	154	204	254	3/8 (.375)	5/16-18	1.688
5	55	105	155	205	255	7/16 (.4375)	7/16-14	2.125
6	56	106	156	206	256	1/2 (.500)	7/16-14	2.125
7	57	107	157	207	257	9/16 (.5625)	1/2-13	2.688
8	58	108	158	208	258	5/8 (.625)	1/2-13	2.688
9	59	109	159	209	259	3/4 (.750)	1/2-13	2.812
10	60	110	160	210	260	7/8 (.875)	9/16-12	3.250
11	61	111	161	211	261	1 (1.00)	3/4-10	3.562
12	62	112	162	212	262	1-1/8 (1.125)	3/4-10	4.000
13	63	113	163	213	263	1-1/4 (1.250)	3/4-10	4.250
14	64	114	164	214	264	1-3/8 (1.375)	7/8-9	5.438
15	65	115	165	215	265	1-1/2 (1.500)	1-8	5.438

Dash number						W	Minimum number of clamps recommended	
Hot dip galvanized		Electrodeposited		Mechanically deposited		Width		
Class 1	Class 2	Class 1	Class 2	Class 1	Class 2	Minimum	Class 1	Class 2
1	51	101	151	201	251	0.938	2	2
2	52	102	152	202	252	0.938	2	2
3	53	103	153	203	253	1.062	2	2
4	54	104	154	204	254	1.062	2	2
5	55	105	155	205	255	1.188	3	2
6	56	106	156	206	256	1.250	3	3
7	57	107	157	207	257	1.375	3	3
8	58	108	158	208	258	1.500	3	3
9	59	109	159	209	259	1.750	3	2
10	60	110	160	210	260	2.125	4	3
11	61	111	161	211	261	2.250	5	4
12	62	112	162	212	262	2.312	5	4
13	63	113	163	213	263	2.500	6	5
14	64	114	164	214	264	2.750	6	5
15	65	115	165	215	265	3.000	7	5

1/ Dimensions are in inches.
2/ Metric equivalents are given for information only.
3/ All dimensions are “reference” dimensions.

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.188	4.77	.625	15.87	1.188	30.17	1.750	44.45	2.812	71.42
.250	6.35	.750	19.05	1.250	31.75	2.125	53.97	3.000	76.20
.313	7.95	.875	22.22	1.281	32.53	2.250	57.15	3.250	82.55
.375	9.52	.938	23.82	1.375	34.92	2.312	58.72	3.562	90.47
.438	11.12	1.000	25.40	1.438	36.52	2.500	63.50	4.000	101.60
.500	12.70	1.062	26.97	1.500	38.10	2.688	68.27	4.250	107.95
.562	14.28	1.125	28.57	1.688	42.87	2.750	69.85	5.438	138.12

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TABLE II. Clamp type III, class 1 weight minimum in lbs. 1/

Type III Class 1			
Clamp size		Weight minimum	
Inches	mm	lbs.	kg
3/16 (.188)	4.77	.21	0.0952
1/4 (.250)	6.35	.21	0.0952
5/16 (.3125)	7.95	.26	0.1179
3/8 (.375)	9.52	.37	0.1678
7/16 (.4375)	11.12	.48	0.2177
1/2 (.500)	12.70	.60	0.2721
9/16 (.5625)	14.28	1.08	0.4898
5/8 (.625)	15.87	1.08	0.4898
3/4 (.750)	19.05	1.34	0.6078
7/8 (.875)	22.22	2.20	0.9979
1 (1.00)	25.40	2.68	1.2156
1-1/8 (1.125)	28.57	2.96	1.3426
1-1/4 (1.250)	31.75	4.03	1.8279
1-3/8 (1.375)	34.92	6.58	2.9846
1-1/2 (1.500)	38.10	6.58	2.9846

1/ Metric equivalents are given for information only.

REQUIREMENTS:

Clamps shall be in accordance with figure 1, figure 2, and table I and table II.

Material steel, carbon shall be in accordance with type III, class I and type III, class 2 of FF-C-450. Bolts of class 2 clamps shall be equivalent to SAE-J429 or ASTM-A449 grade 5 steel or better.

Protective coating:

Finish Code	Finish
A	Protective coating shall be, mechanically deposited zinc coating in accordance with ASTM-B695, class 50, type II, or zinc coated (hot dip galvanized) in accordance with ASTM-A153/A153M or zinc coated (electrodeposited) in accordance with ASTM-B633, Fe/Zn 5, type VI.
C	Cadmium plated, in accordance with SAE-AMS-QQ-P-416, type II, class 3.
Z1	Zinc coating, hot-dip, in accordance with ASTM-A153/A153M
Z2	Zinc coating (electrodeposited) in accordance with ASTM-B633, Fe/Zn 5, type VI.(protects to 120hr in salt spray in accordance with ASTM-B117 and environmentally friendly)
Z3	Zinc coating, mechanically deposited, in accordance with ASTM-B695, class 50, type II (Chromate)
Z4	Zinc coating, mechanically deposited, in accordance with ASTM-B695, class 50, type I (Plain zinc; lowest cost protection, but may be environmentally friendly)

Environmental safety note:

If at all possible and for a safer environment, do not use cadmium or chromate conversion. Recommend to use “Z2” ASTM-B633, Fe/Zn 5, type VI.

Threads shall be in accordance with FF-C-450 specification.

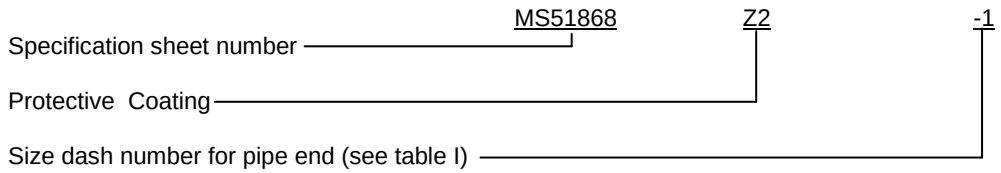
Marking shall be in accordance with FF-C-450 specification.

For design feature purposes, this specification takes precedence over acquisition documents referenced herein.

In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

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Part or Identifying Number (PIN) example.



Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to FF-C-450, this document references the following:

ASTM-A153/A153M
ASTM-A449
ASTM-B117
ASTM-B633
ASTM-B695
SAE-AMS-QQ-P-416
SAE-J429

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – SH
Air Force – 99
DLA – CC

Preparing activity:
DLA – CC

(Project 4030-2012-005)

Review activities:
Army – AT, AV, CR4, GL
Navy – MC, SA, YD
Air Force – 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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